

Flutter Container Decoration

In Flutter, the **Container** widget is used to design UI elements like boxes. To style a Container, we use the **decoration** property with **BoxDecoration**.

Basic Syntax

```
Container(  
  decoration: BoxDecoration(  
    // styling properties  
  ),  
)
```

1. Color

Used to give background color to the container.

```
Container(  
  height: 100,  
  width: 100,  
  color: Colors.blue,  
)
```

Note:

If you use decoration, then **do not use color directly**. Instead, use color inside BoxDecoration.

```
Container(  
  decoration: BoxDecoration(  
    color: Colors.green,  
  ),  
)
```

2. Padding

Padding means **space inside the container**, between content and border.

```
Container(  
  padding: EdgeInsets.all(16),  
  color: Colors.orange,  
  child: Text("Hello Flutter"),  
)
```

Padding Types

```
EdgeInsets.all(10)           // all sides  
EdgeInsets.symmetric(  
  horizontal: 20,  
  vertical: 10,  
)  
EdgeInsets.only(  
  left: 10,  
  top: 5,  
)
```

3. Margin

Margin means **space outside the container**.

```
Container(  
  margin: EdgeInsets.all(20),  
  color: Colors.red,  
  child: Text("Margin Example"),  
)
```

4. Border

Used to add border around the container.

```
Container(  
  decoration: BoxDecoration(  
    border: Border.all(  
      color: Colors.black,  
      width: 2,  
    ),  
  ),  
)
```

5. BorderRadius

Used to make **rounded corners**.

```
Container(  
  decoration: BoxDecoration(  
    color: Colors.blue,  
    borderRadius: BorderRadius.circular(20),  
  ),  
)
```

Different corner radius

```
borderRadius: BorderRadius.only(  
  topLeft: Radius.circular(20),  
  bottomRight: Radius.circular(20),  
)
```

6. BoxShadow

Used to give **shadow effect**.

```
Container(  
  decoration: BoxDecoration(  
    color: Colors.white,  
    boxShadow: [  
      BoxShadow(  
        color: Colors.grey,  
        blurRadius: 10,  
        offset: Offset(4, 4),  
      ),  
    ],  
)
```

```
    ),  
  )  
)
```

7. Gradient

Used for **multi-color background**.

```
Container(  
  decoration: BoxDecoration(  
    gradient: LinearGradient(  
      colors: [Colors.blue, Colors.purple],  
    ),  
  ),  
)
```

Types of Gradients

- LinearGradient
- RadialGradient
- SweepGradient

8. Shape

Used to make container **circle**.

```
Container(  
  width: 100,  
  height: 100,  
  decoration: BoxDecoration(  
    color: Colors.green,  
    shape: BoxShape.circle,  
  ),  
)
```

Expanded Widget in Flutter

The **Expanded widget** is used inside **Row, Column, or Flex**.
It helps a widget **take available free space**.

Why Expanded is Needed?

Without Expanded, widgets take **only required space**.
With Expanded, widgets **share remaining space**.

Basic Example (Row)

```
Row(  
  children: [  
    Expanded(  
      child: Container(  
        height: 100,  
        color: Colors.red,  
      ),  
    ),  
  ],  
)
```

```

Expanded(
  child: Container(
    height: 100,
    color: Colors.blue,
  ),
),
],
)

```

Both containers take **equal width**.

Expanded with flex Property

`flex` decides **how much space** a widget will take.

```

Row(
  children: [
    Expanded(
      flex: 2,
      child: Container(
        color: Colors.green,
        height: 100,
      ),
    ),
    Expanded(
      flex: 1,
      child: Container(
        color: Colors.yellow,
        height: 100,
      ),
    ),
  ],
)

```

Green takes **2 parts**, Yellow takes **1 part**.

Expanded in Column

```

Column(
  children: [
    Expanded(
      child: Container(color: Colors.orange),
    ),
    Expanded(
      child: Container(color: Colors.pink),
    ),
  ],
)

```

Both containers take **equal height**.

Important Notes for Students

- Expanded works **only inside Row, Column, Flex**
- Cannot use Expanded inside `ListView`
- Use `flex` to control space ratio

- Container is for **design**
- Expanded is for **layout**

Padding and Margin in Flutter Widgets

Padding in Flutter

Padding means **space inside a widget**, between the widget's content and its boundary.

Using Padding Widget

```
Padding(
  padding: EdgeInsets.all(16),
  child: Text("Hello Flutter"),
)
```

Padding inside Container

```
Container(
  padding: EdgeInsets.symmetric(horizontal: 20, vertical: 10),
  color: Colors.blue,
  child: Text(
    "Text with Padding",
    style: TextStyle(color: Colors.white),
  ),
)
```

Types of EdgeInsets

```
EdgeInsets.all(10)
EdgeInsets.symmetric(horizontal: 20, vertical: 10)
EdgeInsets.only(left: 10, top: 5)
```

Margin in Flutter

Margin means **space outside a widget**, between widgets.

```
Container(
  margin: EdgeInsets.all(20),
  color: Colors.red,
  child: Text("Margin Example"),
)
```

Note:

Padding → inside space

Margin → outside space

ListTile Widget in Flutter

`ListTile` is a **ready-made widget** used to create **rows in a list** like:

- Contact list

- Settings menu
- Chat list

Basic ListTile Example

```
ListTile(
  title: Text("Neeraj"),
  subtitle: Text("Flutter Developer"),
  leading: Icon(Icons.person),
  trailing: Icon(Icons.arrow_forward),
)
```

Important ListTile Properties

Property	Use
leading	Icon or image at start
title	Main text
subtitle	Secondary text
trailing	Icon at end
onTap	Click action

ListTile with onTap

```
ListTile(
  title: Text("Settings"),
  leading: Icon(Icons.settings),
  onTap: () {
    print("Settings clicked");
  },
)
```

ListTile inside ListView.builder

This is the **most common real-world usage**.

Example: Student List

```
ListView.builder(
  itemCount: 5,
  itemBuilder: (context, index) {
    return ListTile(
      leading: CircleAvatar(
        child: Text("${index + 1}"),
      ),
      title: Text("Student ${index + 1}"),
      subtitle: Text("Flutter Batch"),
      trailing: Icon(Icons.arrow_forward_ios),
    );
  },
)
```

ListTile with Padding & Margin in ListView

```

ListView.builder(
  itemCount: 5,
  itemBuilder: (context, index) {
    return Container(
      margin: EdgeInsets.symmetric(horizontal: 10, vertical: 5),
      decoration: BoxDecoration(
        color: Colors.white,
        borderRadius: BorderRadius.circular(10),
        boxShadow: [
          BoxShadow(color: Colors.grey, blurRadius: 4),
        ],
      ),
      child: ListTile(
        contentPadding: EdgeInsets.all(15),
        leading: Icon(Icons.person),
        title: Text("User ${index + 1}"),
        subtitle: Text("Mobile Developer"),
        trailing: Icon(Icons.call),
      ),
    );
  },
);

```

Important Notes

- **Padding Widget** → used when no Container needed
- **Container padding** → used with decoration
- **ListTile** → pre-designed row widget
- **ListView.builder** → best for large lists
- **ListTile + ListView.builder** → very common interview topic